The documentation process conversion measures necessary to comply with this revision shall be completed by 6 May 1998

INCH-POUND

MIL-PRF-19500/606A
6 February 1998
SUPERSEDING
MIL-S-19500/606
10 November 1992

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED (TOTAL DOSE ONLY)
TRANSISTORS, N-CHANNEL, SILICON TYPES 2N7291, 2N7293,2N7295, AND 2N7297
JANTXVM, D, R, AND JANSM, D AND R

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for an N-Channel, enhancement-mode, MOSFET, radiation hardened (total dose only), power transistor. Two levels of product assurance are provided for each device type specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1, TO-204AA and TO-204AE (similar to TO - 3).

1.3 Maximum ratings. Unless otherwise specified, T_A= +25°C.

Type	P_T 1/ T_C = +25°C	P_T T_A = +25°C	V_{DS}	V_{DG}	V_{GS}	I_{D1} 2/ T_C = +25°C	I_{D2} T_C = +100°C	I_S 2/	I_{DM}	T_J and T_{STG}	V_{ISO} 70,000 feet altitude
	W	W	V dc	V dc	V dc	A dc	A dc	A dc	A (pk)	°C	V dc
2N7291	150	4	100	100	±20	40.0	25.0	40.0	100	-55	N/A
2N7293			200	200		27.0	17.0	27.0	81	to	N/A
2N7295			250	250		20.0	12.0	20.0	60	+150	N/A
2N7297			500	500		10.0	6.0	10.0	30		500

1/ Derate linearly 1.2 W/°C for T_C > +25°C; $P_T = \frac{T_{JM} - T_C}{R_{\theta JC}}$

2/ $I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \, x \, (R_{DS}(on) \, at \, T_{JM})}}$

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

1.4 Primary electrical characteristics at T_C = 25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 I_D = 1.0 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = 1.0 mA dc	Max I_{DSS1} V_{GS} = 0 V_{DS} = 80 percent of rated V_{DS}	Max $r_{DS(on)}$ 1/ V_{GS} = 10 V dc		$R_{\theta JC}$ max	I_{AS} = I_{DM}	E_{AS} at I_{AS}
				T_J = +25°C at I_{D2}	T_J = +125°C at I_{D2}			
	V dc	V dc Min Max	µA dc	ohm	ohm	°C/W	A(pk)	mJ
2N7291	100	2.0 4.0	25	0.055	0.110	0.83	100	500
2N7293	200			0.10	0.220		81	328
2N7295	250			0.17	0.408		60	180
2N7297	500			0.60	1.56		30	45

1/ Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

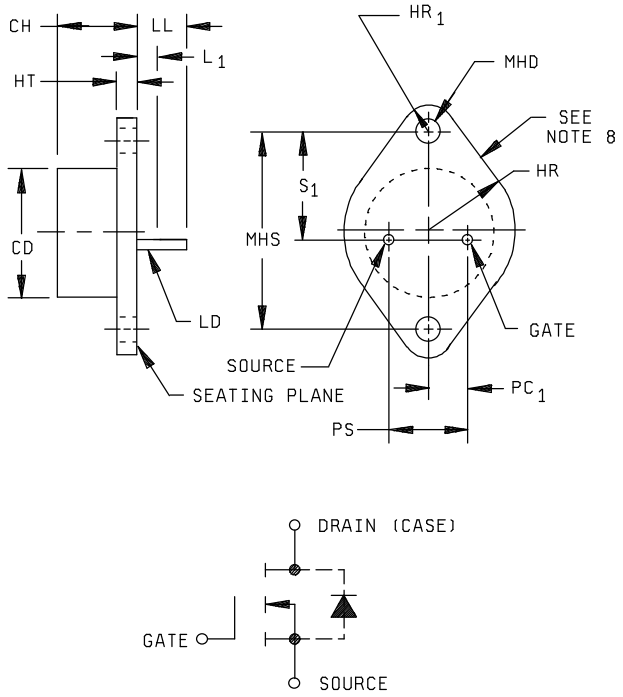
MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.



Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	- - -	.875	- - -	22.23	
CH	.250	.360	6.35	9.15	
HR	.495	.525	12.57	13.34	
HR_1	.131	.188	3.33	4.78	
HT	.060	.135	1.52	3.43	
LD	.057	063	1.45	1.60	5
LD	.038	.043	0.97	1.10	6
LL	.312	.500	7.92	12.70	
L_1	- - -	.050	- - -	1.27	
MHD	.151	.161	3.84	4.09	
MHS	1.117	1.197	29.90	30.40	
PS	.420	.440	10.67	11.18	1, 7
PC_1	.205	.225	5.21	5.72	1, 7
S_1	.655	.675	16.64	17.15	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These dimensions should be measured at points .050 inch (1.27 mm) and .055 inch (1.40 mm) below seating plane. When gauge is not used, measurements will be made at the seating plane.
4. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
5. These dimensions pertain to the 2N7291 and 2N7293 types.
6. These dimensions pertain to the 2N7295 and 2N7297 types.
7. Mounting holes shall be deburred on the seating plane side.
8. Drain is electrically connected to the case.

FIGURE 1. Physical dimensions for TO-204 (similar to TO - 3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

I_{AS} - Rated avalanche current, non-repetitive

3.3 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (TO204) herein.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52; a copper core or plated core is permitted. Lead finish shall be solderable in accordance with MIL-PRF-19500 and herein. Where a choice of lead finish is desired, it shall be specified in the contract or purchase order (see 6.2).

3.3.2 Internal construction. Multiple chip construction is not be permitted to meet the requirements of this specification.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic discharge protection.

3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended (see 3.5).

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100$ k, whenever bias voltage is to be applied drain to source.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4 and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-PRF-19500, and table III herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	MIL-STD-750, method 3161 (see 4.5.3)	MIL-STD-750, method 3161 (see 4.5.3)
1/	Gate stress test (see 4.5.5)	Gate stress test (see 4.5.5)
1/	MIL-STD-750, method 3470, E_{AS} test (see 4.5.4)	MIL-STD-750, method 3470, E_{AS} test (see 4.5.4)
2/	Subgroup 2 of table I herein	Subgroup 2 of table I herein
9	I_{GSS}, I_{DSS1}	Not applicable
10	MIL-STD-750, method 1042, test condition B	MIL-STD-750, method 1042, test condition B
11	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(on)}$, $V_{GS(th)}$ subgroup 2 of table I herein; ΔI_{GSSF1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 10 μA dc or ± 100 percent of initial value, whichever is greater.	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(on)}$, $V_{GS(th)}$ subgroup 2 of table I herein
12	MIL-STD-750, method 1042, test condition A	MIL-STD-750, method 1042, condition A or T_A = +175°C and t = 48 hours min 3/
13	Subgroups 2 and 3 of table I herein; ΔI_{GSSF1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 10 μA dc or ± 100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ± 20 percent of initial value. $\Delta V_{GS(th)}$ = ± 20 percent of initial value.	Subgroups 2 and 3 of table I herein; ΔI_{GSSF1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 10 μA dc or ± 100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ± 20 percent of initial value. $\Delta V_{GS(th)}$ = ± 20 percent of initial value.

1/ Shall be performed any time after screen 10.

2/ Shall be performed after E_{AS} test, method 3161, and gate stress test.

3/ Use of this accelerated screening option requires a 1,000 hour life test in accordance with the applicable group E, subgroup 2 life test, and end-points specified herein to be provided to the qualifying activity for review and acceptance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. Alternate flow is allowed for quality conformance inspection in accordance with appendix E of MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500 and table I herein. (End point electrical measurements shall be in accordance with the applicable steps of table IV herein.)

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIa (JANS) and table VIb (JANTX and JANTXV) of MIL-PRF-19500, and herein. Electrical measurements (endpoints) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.2.1 Group B inspection, appendix E, table VIa of MIL-PRF-19500.

Subgroup	Method	Conditions
B2	2077	
B5	1042	V_{DS} = rated; T_A = +175°C; t = 120 hours min. Read and record $V_{BR(DSS)}$ (pre and post) at I_D = 1 mA. Read and record I_{DSS} (pre and post) in accordance with table IV.
B5	1042	Condition B, V_{GS} = rated; T_A = +175°C, t = 24 hours.
B5	2037	Test condition A, all internal wires for each device shall be pulled separately.
B5	3161	See 4.5.2

4.4.2.2 Group B inspection, appendix E, table VIb of MIL-PRF-19500.

Subgroup	Method	Conditions
B2	1051	Test condition G, 25 cycles.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500, and herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.3.1 Group C inspection, appendix E, table VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	1056	Condition B.
C2	2036	Test condition A, weight = 10 pounds; t = 15 s.
C6	1042	Test condition D, 6,000 cycles; a cycle shall be 30 seconds minimum.

4.4.4 Group D Inspection. Group D inspection shall be conducted in accordance with appendix E, table VIII of MIL-PRF-19500 and table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit of $R_{\theta JC(max)}$ = 0.83°C/W. The following parameter measurements shall apply.

a. Measuring current (I_M) ..... 10 mA

b. Drain heating current (I_H) ..... 4 A

c. Heating time (t_H) ..... Steady-state (see MIL-STD-750, method 3161 for definition)

d. Drain-source heating voltage (V_H) ..... 25 V

e. Measurement time delay (t_{MD}) ..... 30 to 60 μs

f. Sample window time (t_{SW}) ..... 10 μs maximum

4.5.3 Thermal response ΔV_{SD} measurement). The ΔV_{SD} measurement shall be performed in accordance with method 3161 of MIL-STD-750. The ΔV_{SD} conditions (I_H and V_H) and maximum limit shall be derived by each vendor from the thermal response curves (see figure 2) and shall be specified in the certificate of conformance prior to qualification. The following parameter measurements shall apply:

a. Measuring current (I_M) ..... 10 mA

b. Drain heating current (I_H) ..... 4 A minimum

c. Heating time (t_H) ..... 10 ms

d. Drain-source heating voltage (V_H) ..... 25 V

e. Measurement time delay (t_{MD}) ..... 30 to 60 μs

f. Sample window time (t_{SW}) ..... 10 μs maximum

4.5.4 Single pulse avalanche energy (E_{AS}).

a. $I_{AS} = I_{DM}$

b. L = .1 mH

c. $E_{AS} = .5\ LI_{AS}^2$

d. Initial junction temperature = +25°C +10°C, -5°C

4.5.5 Gate stress test.

a. V_{GS} = 30 V minimum.

b. t = 250 μs minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Units
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Breakdown voltage, drain to source	3407	V_{GS} = 0 V; I_D = 1 mA dc, bias condition C	$V_{(BR)DSS}$			
2N7291				100		V dc
2N7293				200		V dc
2N7295				250		V dc
2N7297				500		V dc
Gate to source voltage (threshold)	3403	$V_{DG} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(th)1}$	2.0	4.0	V dc
Gate current	3411	V_{GS} = +20 and -20 V dc, bias condition C, V_{DS} = 0	I_{GSS1}		± 100	nA dc
Drain current	3413	V_{GS} = 0 V dc, bias condition C V_{DS} = 80 percent of rated V_{DS}	I_{DSS1}		25	µA dc
Static drain to source “ON”-state resistance	3421	V_{GS} = 10 V dc, condition A, pulsed (see 4.5.1), $I_d = I_{d2}$	$r_{DS(on)1}$			
2N7291					0.055	Ω
2N7293					0.10	Ω
2N7295					0.17	Ω
2N7297					0.60	Ω
Static drain to source “ON”-state resistance	3421	V_{GS} = 10 V dc, condition A, pulsed (see 4.5.1), $I_d = I_{d1}$	$r_{DS(on)2}$			
2N7291					0.058	Ω
2N7293					0.105	Ω
2N7295					0.179	Ω
2N7297					0.630	Ω
Forward voltage	4011	Pulsed (see 4.5.1), Id = Id1 V_{GS} = 0 V dc	V_{SD}			
2N7291					1.8	V dc
2N7293					1.8	V dc
2N7295					1.8	V dc
2N7297					1.8	V dc

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Units
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		$T_C = T_J = +125°C$				
Gate current	3411	V_{GS} = +20 and -20 V dc, bias condition C, V_{DS} = 0	I_{GSS2}		± 200	nA dc
Drain current	3413	V_{GS} = 0 V dc, bias condition C V_{DS} = 100 percent of rated V_{DS}	I_{DSS2}		1.0	mA dc
		V_{DS} = 80 percent of rated V_{DS}	I_{DSS2}		0.25	mA dc
Static drain to source "ON"-state resistance	3421	V_{GS} = 10 V dc, pulsed (see 4.5.1), $I_d = I_{d2}$	$r_{DS(on)3}$			
2N7291					0.110	Ω
2N7293					0.220	Ω
2N7295					0.408	Ω
2N7297					1.56	Ω
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(th)2}$	1.0		V dc
Low temperature operation:		$T_C = T_J = -55°C$				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(th)3}$		5.0	V dc
Subgroup 4						
Switching time test	3472	$I_D = I_{D1}$, V_{GS} = 10 V dc, R_G = 25Ω, V_{DD} = 50 percent of rated V_{DS}				
Turn-on delay time			$t_{d(on)}$			
2N7291					170	ns
2N7293					170	ns
2N7295					150	ns
2N7297					160	ns
Rise time			t_r			
2N7291					1,120	ns
2N7293					600	ns
2N7295					450	ns
2N7297					260	ns

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Units
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Turn-off delay time			$t_{d(off)}$			
2N7291					420	ns
2N7293					580	ns
2N7295					650	ns
2N7297					750	ns
Fall time			t_f			
2N7291					380	ns
2N7293					300	ns
2N7295					260	ns
2N7297					180	ns
Subgroup 5						
Safe operating area test	3474	See figure 3 t_p = 10 ms minimum V_{DS} = 80 percent of max rated V_{DS} ($V_{DS} \leq 200$)				
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7				
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition B				
On-state gate charge		V_{DD} = 0.5 BV_{DSS}, $I_D = I_{D1}$ $V_{GS} \leq 20V$, $I_{GS1} = I_{GS2}$	$Q_{g(on)}$			
2N7291					230	nC
2N7293					244	nC
2N7295					252	nC
2N7297					258	nC

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Units
	Method	Conditions		Min	Max	
Subgroup 7 - Continued						
Gate to source charge			Q_{gs}			
2N7291					63	nC
2N7293					53	nC
2N7295					48	nC
2N7297					34	nC
Gate to drain charge			Q_{gd}			
2N7291					123	nC
2N7293					125	nC
2N7295					126	nC
2N7297					123	nC
Reverse recovery time	3473	di/dt = 100 A/µs, $V_{DD} \leq 30$ V, $I_d = I_{d1}$	T_{rr}			
2N7291					1,400	ns
2N7293					1,700	ns
2N7295					2,000	ns
2N7297					2,300	ns

1/ For sampling plan, see MIL-PRF-19500.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

TABLE II. Group D inspection.

Inspection 1/	MIL-STD-750		Symbol	Preirradiation limits		Preirradiation limits		Units
	Method	Conditions		M, D, and R		M, D, and R		
				Min	Max	Min	Max	
Subgroup 1								
Not applicable								
Subgroup 2		T_C = +25°C						
Steady state total dose irradiation	1019	2/ 3/						
End point electricals								
Breakdown voltage, drain to source	3407	V_{GS} = 0; I_D = 1 mA bias condition C	$V_{BR(DSS)}$					
2N7291				100		100		V dc
2N7293				200		200		V dc
2N7295				250		250		V dc
2N7297				500		500		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = 1 mA	$V_{GS(th)1}$					
2N7291				2.0	4.0	2.0	4.0	V dc
2N7293				2.0	4.0	2.0	4.0	V dc
2N7295				2.0	4.0	2.0	4.0	V dc
2N7297				2.0	4.0	2.0	4.0	V dc
Gate current	3411	V_{GS} = 20 V; V_{DS} = 0 bias condition C	I_{GSSF1}		100		100	nA dc
Gate current	3411	V_{GS} = 20 V; V_{DS} = 0 bias condition C	I_{GSSR1}		-100		-100	nA dc
Drain current	3413	V_{GS} = 0 bias condition C V_{DS} = 80 percent of rated V_{DS} (preirradiation)	I_{DSS1}					
2N7291					25		25	µA dc
2N7293					25		25	µA dc
2N7295					25		25	µA dc
2N7297					25		25	µA dc

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

TABLE II. Group D inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Preirradiation limits		Preirradiation limits		Units
	Method	Conditions		M, D, and R		M, D, and R		
				Min	Max	Min	Max	
Subgroup 2 - Continued								
Static drain to source on-state resistance	3421	V_{GS} = 10 V; condition A pulsed (see 4.5.1) $I_D = I_{D2}$	$R_{DS(on)1}$					
2N7291					0.055		0.055	Ω
2N7293					0.100		0.100	Ω
2N7295					0.170		0.170	Ω
2N7297					0.600		0.600	Ω
		V_{GS} = 10 V, condition A						
2N7291								Ω
2N7293								Ω
2N7295								Ω
2N7297								Ω
Drain source on-state voltage	3405	V_{GS} = 10 V; $I_D = I_{D1}$; condition A; pulsed, see 4.5.1 $I_D = I_{D1}$	$V_{DS(on)}$					
2N7291					2.32		2.32	V dc
2N7293					2.70		2.70	V dc
2N7295					3.58		3.58	V dc
2N7297					6.30		6.30	V dc
		V_{GS} = 10 V Condition A						
2N7291								V dc
2N7293								V dc
2N7295								V dc
2N7297								V dc

1/ For sampling plan, see MIL-PRF-19500.

2/ Inspection requires all subgroup 2 (group D) measurements after exposure to both of the following in situ bias conditions:

V_{GS} = 10 V; V_{DS} = 0

V_{GS} = 0 V; V_{DS} = 80 percent of rated V_{DS}

Each bias condition requires a separate total dose sample.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

TABLE III. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Qualification and large lot quality conformance inspection
	Method	Conditions	
Subgroup 1			12 devices c = 0
Temperature cycling (air to air)	1051	-55°C to +150°C, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7	
Subgroup 2 1/			12 devices c = 0
Steady-state reverse bias	1042	Condition A, 1,000 hours	
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7	
Steady-state reverse bias	1042	Condition B, 1,000 hours.	
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7	
Subgroup 3			
Not applicable			
Subgroup 4			12 devices c = 0
Thermal resistance	3161	$R_{\theta JC}$ = 0.83° C/W maximum. See 4.5.2	

1/ A separate sample for each test shall be pulled.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

TABLE IV. Groups A, B, and C electrical measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Units
		Method	Conditions		Min	Max	
1.	Breakdown voltage drain to source	3407	V_{GS} = 0 V; I_D = 1 mA dc; bias condition C	$V_{(BR)DSS}$			
	2N7291				100		V dc
	2N7293				200		V dc
	2N7295				250		V dc
	2N7297				500		V dc
2.	Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$; I_D = 1 mA dc	$V_{GS(th)1}$	2.0	4.0	V dc
3.	Gate current	3411	V_{GS} = +20 and -20 V dc; V_{DS} = 0 V; bias condition C	I_{GSS1}		± 100	nA dc
4.	Drain current	3413	V_{GS} = 0 V dc; bias condition C, V_{DS} = 80 percent of rated V_{DS}	I_{DSS1}		25	µA dc
5.	Static drain to source “ON”-state resistance	3421	V_{GS} = 10 V dc; condition A pulsed (see 4.5.1), $I_D = I_{D2}$	$r_{DS(on)1}$			
	2N7291					0.055	Ω
	2N7293					0.100	Ω
	2N7295					0.170	Ω
	2N7297					0.600	Ω
6.	Static drain to source “ON”-state resistance	3421	V_{GS} = 10 V dc; condition A pulsed (see 4.5.1), $I_D = I_{D1}$	$r_{DS(on)2}$			
	2N7291					0.058	Ω
	2N7293					0.105	Ω
	2N7295					0.179	Ω
	2N7297					0.630	Ω
7.	Forward-voltage (source drain diode)	4011	Pulsed (see 4.5.1) V_{GS} = 0 V dc, $I_D = I_{D1}$	V_{SD}			
	2N7291					1.8	V dc
	2N7293					1.8	V dc
	2N7295					1.8	V dc
	2N7297					1.8	V dc
8.	Thermal response	3161	See 4.5.3	ΔV_{SD}			

1/ The electrical measurements for appendix E, table VIa (JANS) of MIL-PRF-19500 are as follows:

a. Subgroup 3, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.

b. Subgroup 4, see table IV herein, steps 1, 2, 3, 4, 5, 6, 7, and 8.

c. Subgroup 5, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

TABLE IV. Groups A, B, and C electrical measurements. Continued.

2/ The electrical measurements for appendix E, table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.

b. Subgroup 3 see table IV herein, steps 1, 2, 3, 4, 5, 6, 7, and 8.

3/ The electrical measurements for appendix E, table VII of MIL-PRF-19500 are as follows:

Subgroups 2, 3, and 6, see table IV herein, steps 1, 2, 3, 4, 5, 6, and 7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

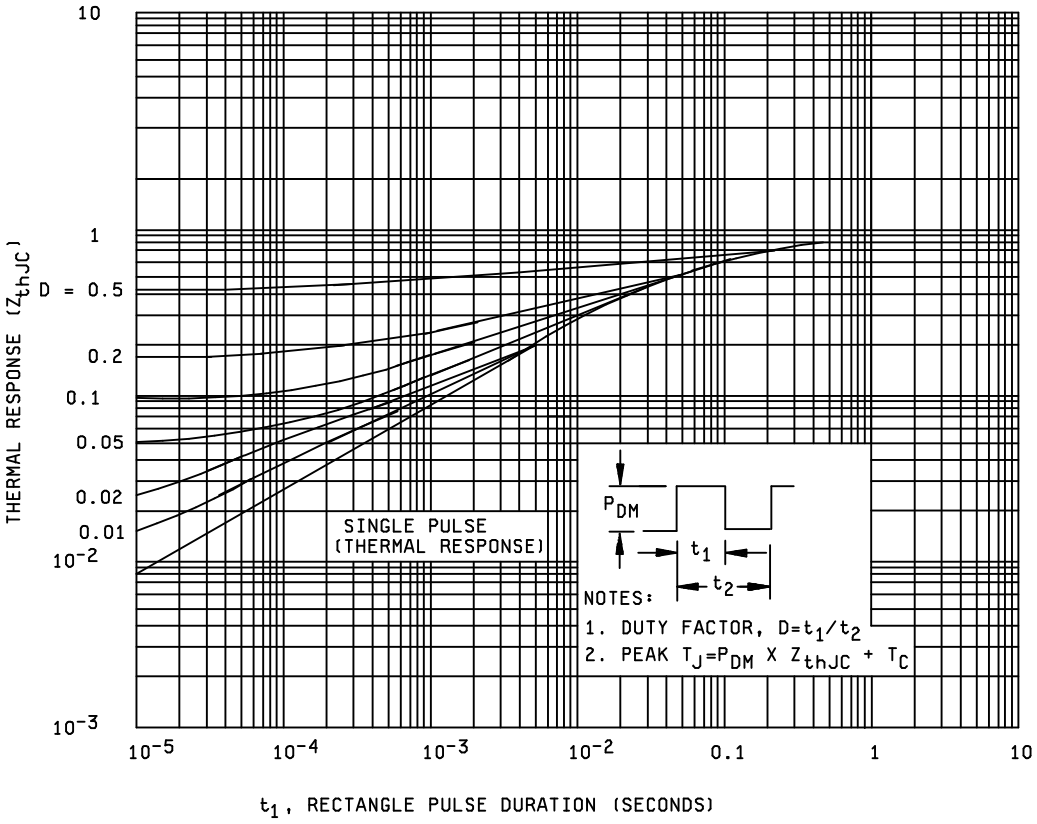


FIGURE 2. Thermal response curves.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

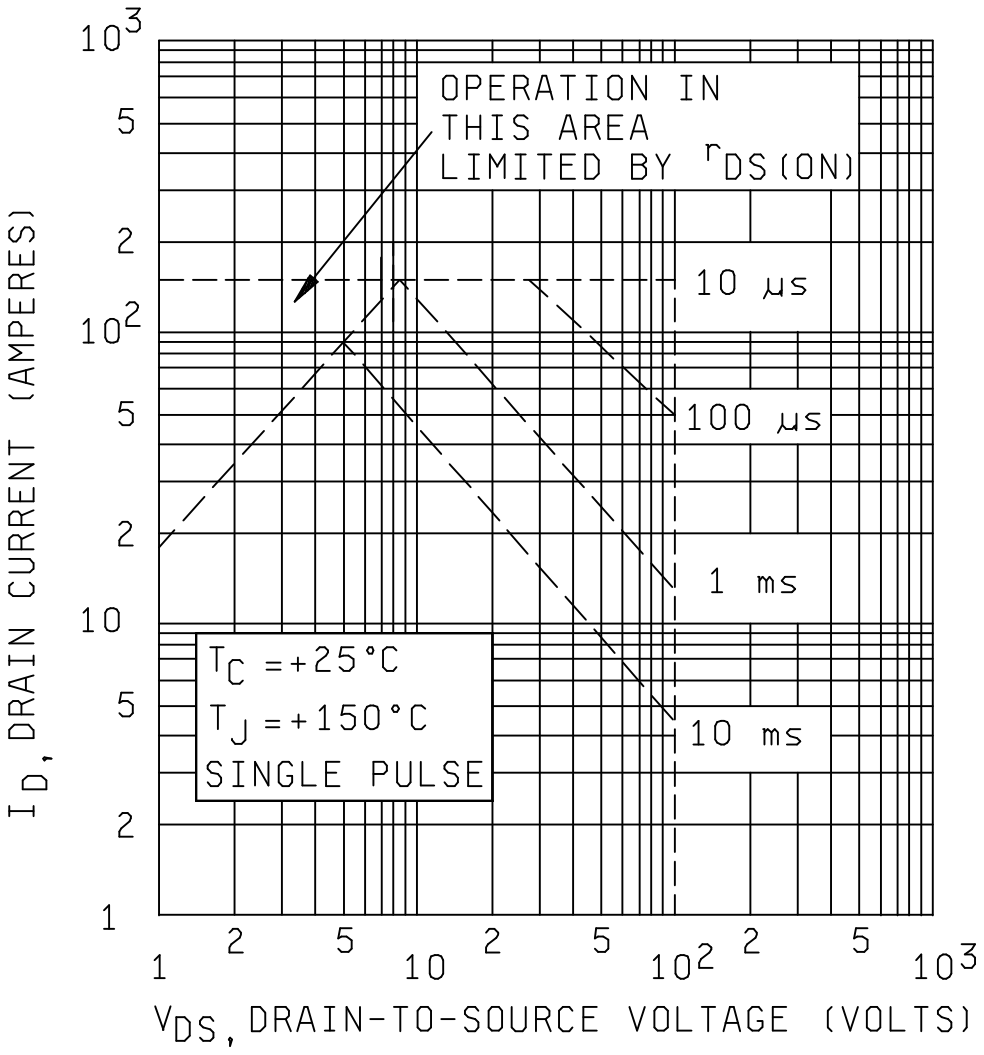


FIGURE 3. Safe operating area graphs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

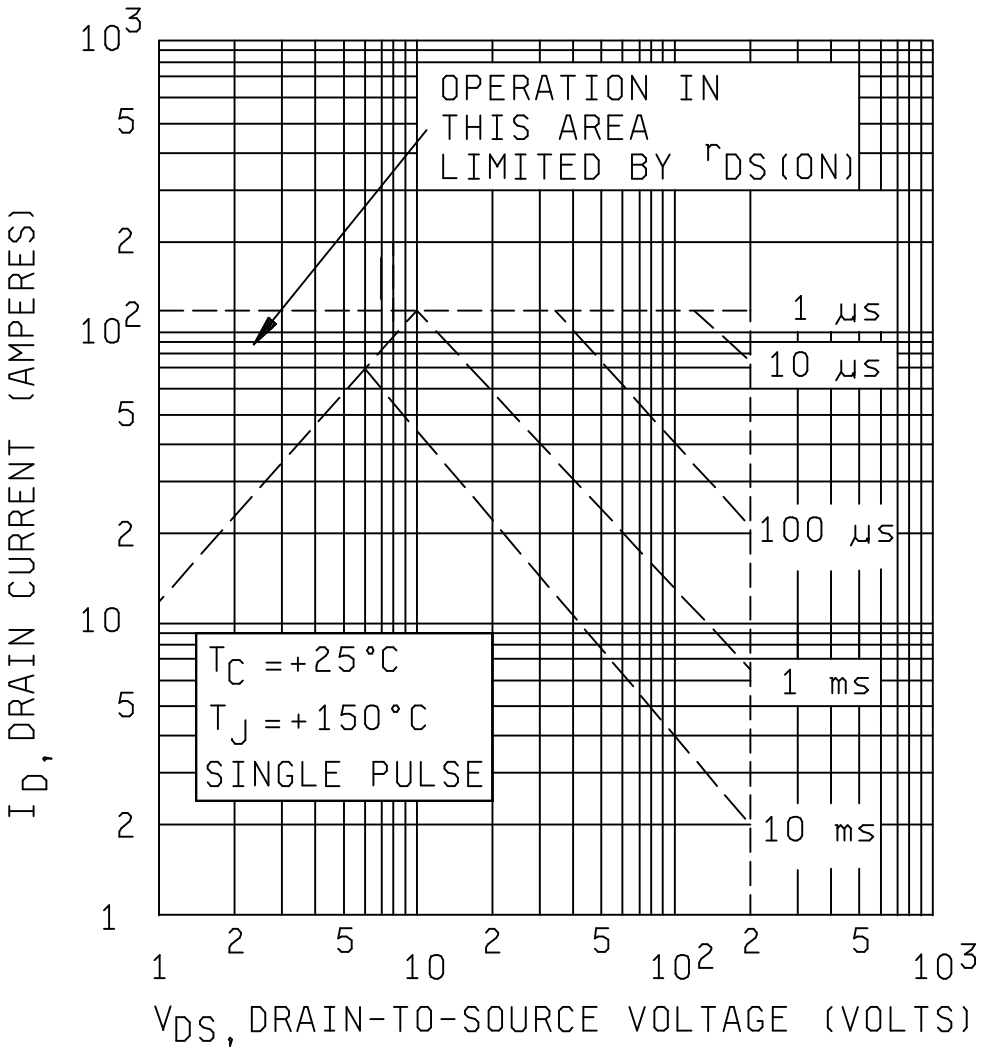


FIGURE 3. Safe operating area graphs - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

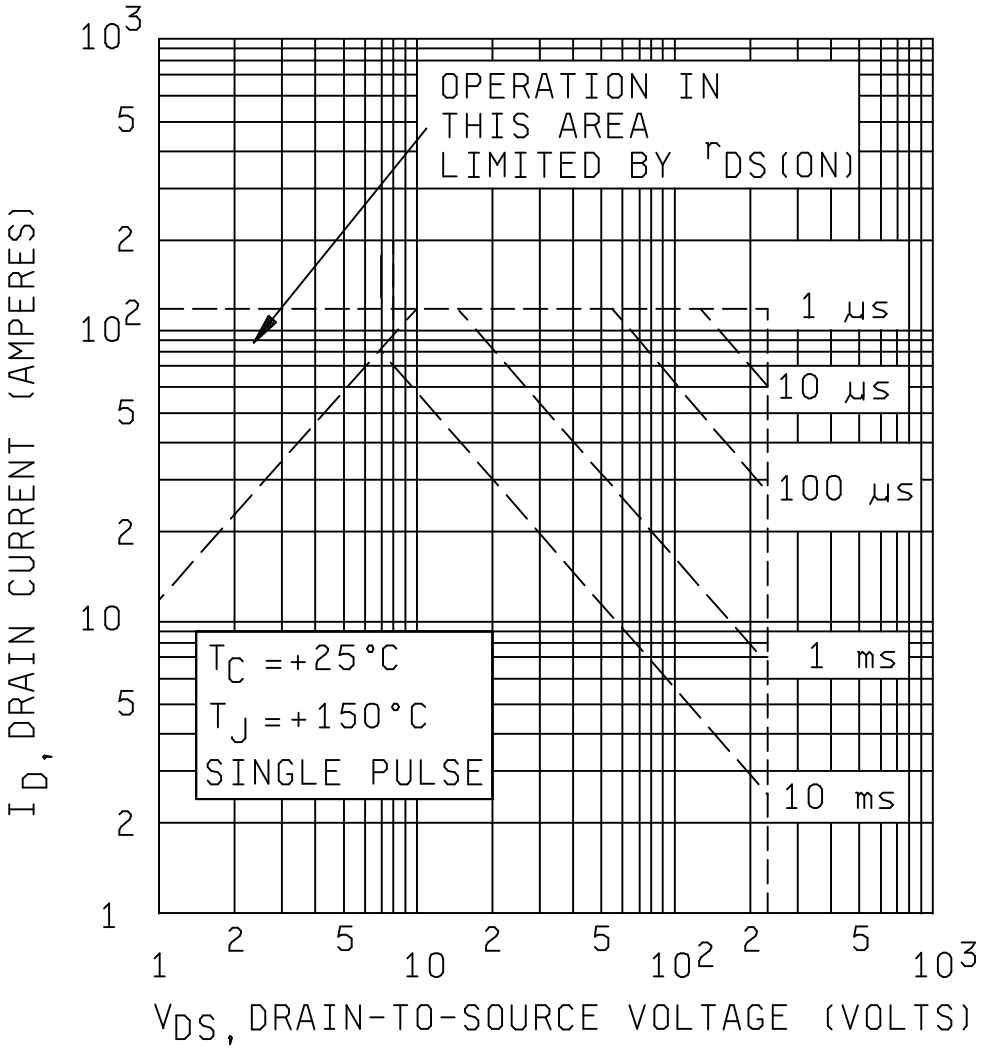


FIGURE 3. Safe operating area graphs - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

2N7297

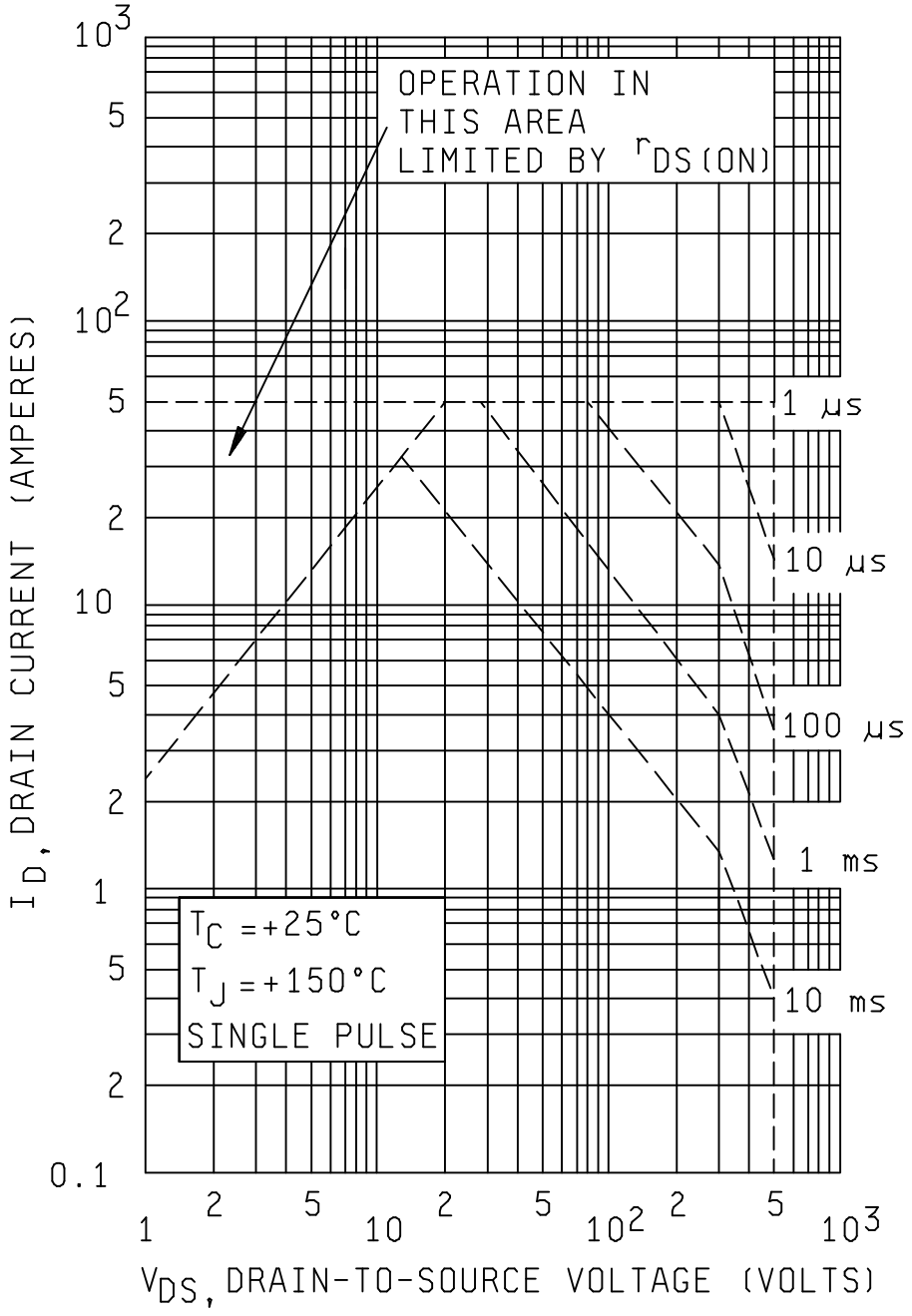


FIGURE 3. Safe operating area graphs - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's Part or Identifying Number (PIN). This information in no way implies that manufacturer's PIN's are suitable for the military PIN.

Preferred types	Commercial types
2N7291	FRK150 1/
2N7293	FRK250 1/
2N7295	FRK254 1/
2N7297	FRK450 2/

1/ FRKxxxM, FRKxxxD, FRKxxxR, 3 k, 10 k, 100 k RAD(Si).
2/ FRM450M, FRM450D, FRM450R, 3 k, 10 k, 100 k RAD(Si).

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Navy - TD
Air Force - 19, 70, 80

Preparing activity:
DLA - CC

(Project 5961-1902-03)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/606A	**2. DOCUMENT DATE (YYMMDD)** 980206

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED (TOTAL DOSE ONLY) TRANSISTORS, N-CHANNEL, SILICON TYPES 2N7291, 2N7293,2N7295, AND 2N7297 JANTXVM, D, R, AND JANSM, D AND R

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. Point of contact: Alan Barone	b. TELEPHONE Commercial 614-692-0510 DSN 850-0510 FAX 614-692-6939 EMAIL alan_barone@dscc.dla.mil
c. ADDRESS: Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:02Z
Check the source to verify that this is the current version before use.